Production and Optimization of Biodiesel from Palm Kernel Oil (PKO) Using Ash Coconut Husk as Catalyst

By

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DEDICATION

This project is dedicated to my parents may almighty be pleased with them. Aaameen

CERTIFICATION

This research work titled ‘‘Production and Optimization of Biodiesel from Palm Kernel Oil (PKO) Using Ash Coconut Husk as Catalyst” by Abdulaziz Sani Adm No: 20133045 has meet the requirements for the award of Bachelor of Science Degree in Chemistry of Sokoto State University, Sokoto and it is approved for its contribution to knowledge.

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Abstract

This study examined the production and optimization of biodiesel from palm kernel oil (PKO) using ash derived from coconut husk as a low-cost, eco-friendly catalyst. The research aimed to provide a sustainable alternative to fossil fuels by utilizing abundant agricultural wastes for both oil and catalyst sources. Palm kernel oil was extracted using a Soxhlet apparatus, while the catalyst was prepared by calcining coconut husk at 500 °C. Transesterification was carried out with methanol, and process parameters such as methanol-to-oil ratio, catalyst loading, and reaction time were optimized using the Box–Behnken design under Response Surface Methodology (RSM). The optimized reaction yielded up to 92.91 % biodiesel, with catalyst loading and methanol-to-oil ratio identified as the most significant factors affecting conversion efficiency. The produced biodiesel met ASTM D6751 standards, showing a flash point of 160 °C, viscosity of 3.90 mm^2/s, and moisture content of 0.02 wt %. Although cold flow properties were slightly high due to the saturated nature of PKO, overall fuel quality was excellent. In conclusion, this study demonstrated that palm kernel oil and coconut husk ash can be effectively utilized to produce high-quality biodiesel through an environmentally friendly process. Further research should focus on improving catalyst reusability, enhancing cold flow behavior, and scaling up production for industrial application.

CHAPTER ONE

1.0 INTRODUCTION AND LITERATURE REVIEW

1.1 Background of study

The growing global demand for sustainable and eco-friendly energy sources has intensified research into alternative fuels, particularly biodiesel. Biodiesel, a renewable, biodegradable, and non-toxic fuel, is typically derived from vegetable oils, animal fats, or recycled cooking oils through a chemical process known as transesterification. However, concerns over feedstock cost, food-versus-fuel competition, and reliance on chemical catalysts have driven interest in using some waste-derived resources for both oil and catalysts [1].

Biodiesel is a renewable, biodegradable fuel that is chemically produced from natural oils or fats, such as vegetable oil, animal fat, or waste cooking oil, through a process called transesterification. In this process, the triglycerides in the oils or fats react with an alcohol (usually methanol or ethanol) in the presence of a catalyst such as ash Coconut Husk or potassium hydroxide, to produce fatty acid methyl esters (FAME) which is biodiesel and glycerol as a byproduct [2]. Biodiesel is widely recognized as one of the most promising alternative fuels to petroleum diesel. It can be utilized in conventional diesel engines without requiring any modifications, making it a convenient and practical substitute. Derived from renewable vegetable oils, biodiesel is a clean-burning, environmentally friendly energy source that contributes to reduced greenhouse gas emissions and improved air quality [2].

The palm kernel oil (*Elaeis guineensis*) is a highly productive tropical crop cultivated extensively in Africa, Asia, and South America. It yields two distinct types of oils: palm oil, extracted from the fleshy mesocarp, and palm kernel oil (PKO), obtained from the seed or kernel of the fruit. Palm kernel oil has emerged as an important renewable feedstock for

biodiesel production, particularly in tropical regions where the crop is abundant and economically significant [3].

Chemically, palm kernel oil is rich in medium-chain saturated fatty acids, with lauric acid (C12:0) comprising approximately 45–55% of its fatty acid profile, followed by myristic acid (C14:0), capric acid (C10:0), and smaller amounts of oleic and linoleic acids. This composition confers high oxidative stability to the oil, making it an excellent candidate for biodiesel production with a long shelf life and reduced susceptibility to rancidity [4].

Physiochemically, palm kernel oil has a relatively low iodine value (typically 13–23 g I_2/100 g), a high saponification value (~240–260 mg KOH/g), and moderate viscosity (~32–36 mm^2/s at 40 °C). These properties support efficient transesterification reactions, yielding biodiesel that meets established standards. The triglyceride structure of PKO reacts readily with methanol or ethanol in the presence of catalysts such as NaOH, KOH, or heterogeneous materials, producing fatty acid methyl esters (biodiesel) and glycerol as a byproduct [4].

The use of palm kernel oil as a biodiesel feedstock offers several advantages. It is widely available, cost-effective, and renewable, reducing dependence on fossil fuels while supporting rural economies. Its high conversion efficiency and favorable fuel properties make it particularly attractive in regions seeking sustainable energy alternatives. Consequently, the development of biodiesel production from palm kernel oil aligns with global efforts to promote cleaner, more sustainable energy sources [5].

Coconut husk ash (CHA) has been studied as a low-cost, heterogeneous catalyst for biodiesel production. In one study, CHA was prepared by calcining coconut husk and used to trans esterify Cerbera manghas seed oil, achieving a biodiesel yield of 88.6% under optimized conditions (10 wt% catalyst, 6:1 methanol-to-oil ratio, 3-hour reaction) [6]. The catalyst

showed good reusability and ease of separation. However, the study did not apply CHA to palm kernel oil or assess important fuel properties such as cetane number or viscosity. This research seeks to fill that gap by using CHA with palm kernel oil and evaluating both yield and combustion related properties [7].

Equally important is the development of low-cost, environmentally benign catalysts. In this context, ash derived from coconut husks a lignocellulosic biomass waste rich in alkali and alkaline earth metals has emerged as a potential solid base catalyst [8].

This study explores the production of biodiesel from palm kernel oil using ash from coconut husk as a heterogeneous catalyst. The approach aligns with green chemistry principles by valorizing agricultural wastes, reducing environmental impact, and contributing to circular bioeconomy efforts. The investigation focuses on optimizing transesterification conditions, evaluating catalyst efficiency, and assessing the fuel properties of the resulting biodiesel in accordance with international standards [9].

1.2 Literature review

Biodiesel production from Palm Kernel Oil (PKO) is a promising strategy for sustainable fuel generation. Onyia et al. [10]. Demonstrated high-yield biodiesel production from PKO using a heterogeneous MgO catalyst, achieving 93.08 % yield under optimized conditions determined via Response Surface Methodology (RSM). The optimized conditions are oil to methanol ratio

9:1, 3 wt% catalyst loading and 45-minute reaction time. Their study carefully characterized

PKO's physicochemical properties, confirming its suitability as feedstock, and showed that biodiesel produced met international fuel standards.

However, their reliance on commercial MgO highlights a research gap in exploring low-cost, waste-derived catalysts. Using agricultural residues like coconut husk ash offers an environmentally friendly alternative with potential for circular economy integration. Building on

Onyia et al.'s optimization framework [10]. The present study aim to adapt RSM to optimize PKO transesterification using coconut husk ash as a heterogeneous catalyst, advancing both waste valorization and sustainable energy goals [10].

Etim et al. [11] reported the conversion of flax seed oil into biodiesel using a bio-alkaline salt catalyst derived from Musa acuminate (banana) peels. Under optimized reaction conditions of methanol ratio 9:2, 3 wt% catalyst loading, 60-minute reaction time and temperature of 55oC, they achieved an experimental yield of 96.50 wt.%, demonstrating the potential of this approach as a sustainable alternative to fossil fuels.

Similarly, Novita et al. [12]. Successfully converted waste palm cooking oil into biodiesel employing a green, recyclable catalyst produced from palm kernel shell ash. Their process yielded 99.01%, meeting established physicochemical quality standards. Furthermore, the approach offers notable advantages for large-scale implementation by significantly reducing waste generation and production costs.

Arumugam and Sankaranarayanan, [13] study explored sugarcane leaf ash as a catalyst for Calophyllum inophyllum methyl ester production. Through characterization and response surface methodology, optimal conditions of 19:1 methanol to oil ratio, 5 wt% catalyst, and 64°C temperature yield 97% FAME. Reusability data demonstrates 97.85% yield over 6 cycles, and blended B10 and B80 exhibit improved efficiency in backup energy usage.

The study by onyia et al. [10] demonstrates the technical feasibility of producing high-quality biodiesel from palm kernel oil using MgO as a heterogeneous catalyst. The optimized process parameters, combined with compliance with international fuel standards, underscore its potential for industrial-scale adoption. This approach supports circular economy principles and contributes directly to key Sustainable Development Goals (SDGs) targets. Future research should prioritize lifecycle assessments, catalyst durability studies, and commercialization strategies to ensure sustainable and economically viable deployment.

1.3 Problem Statement

The global dependence on fossil fuels presents significant environmental, economic, and sustainability challenges. Petroleum-derived diesel is non-renewable, contributes substantially to greenhouse gas emissions, and faces rising costs due to depleting reserves. Transitioning to biodiesel a renewable, cleaner-burning fuel offers a promising pathway toward sustainable energy solutions [10]. However, the large-scale adoption of biodiesel is constrained by the high cost of conventional production methods, which typically rely on refined vegetable oils and expensive homogeneous chemical catalysts. These factors limit economic feasibility, particularly in developing regions where cost-effective and locally available alternatives are crucial [11]. Simultaneously, substantial quantities of agricultural and food-processing by-products, such as coconut husks and palm kernel residues, are often discarded as waste. These materials represent an underutilized resource with significant potential for valorization in biofuel production. Notably, coconut husk ash, rich in metal oxides, may serve as an inexpensive, environmentally friendly heterogeneous catalyst for the transesterification of oils [14].

Palm kernel oil, abundant in many tropical regions, is a promising, locally sourced feedstock for biodiesel production. Leveraging coconut husk ash as a catalyst offers a sustainable, low-

cost alternative to conventional chemical catalysts while also addressing agricultural waste disposal challenges [15].

1.4 Justification of the study

The need to reduce reliance on fossil fuels and address environmental challenges makes biodiesel production an important research priority. However, conventional biodiesel production often depends on expensive refined vegetable oils and chemical catalysts, making it less affordable in developing regions.

Palm kernel oil is widely available in many tropical countries and offers good properties for biodiesel production. Using it as feedstock supports local agriculture, reduces waste, and provides a renewable, cost-effective alternative to fossil diesel.

At the same time, coconut husk is an abundant agricultural waste that is often discarded. By converting coconut husk ash into a heterogeneous catalyst, this study promotes waste valorization and green chemistry principles. Such catalysts are cheaper, reusable, and more environmentally friendly compared to traditional chemical catalysts. By valorizing coconut husk—an abundant agricultural residue into a functional catalyst, this research supports a circular economy model and encourages low-cost, decentralized biodiesel production. Additionally, the use of locally sourced materials and simple processing techniques promotes technological accessibility, local empowerment, and small-scale energy independence. The outcomes of this study may also provide valuable input for ongoing policy reforms aimed at expanding the renewable energy sector, especially as many African nations seek to diversify their energy portfolios and meet global sustainability targets.

1.5 Aim and Objectives

The primary aim of this study is to develop and optimize a sustainable process for biodiesel production through the transesterification of palm kernel oil, employing calcined coconut

husk ash as a solid base heterogeneous catalyst. This research is positioned within the broader context of renewable energy production, waste valorization, and green chemistry. The specific objectives are to:

i. Extract the oil from paml kernel seed using solvent extraction method
ii. Convert agro-industrial waste (palm kernels and coconut husks) into high-value biodiesel fuel.
iii. To develop, optimize coconut husk ash as a cost-effective, environmentally friendly heterogeneous catalyst, and to determine the optimal transesterification reaction parameters (including catalyst concentration, methanol-to oil ratio, reaction and time) required to maximize biodiesel yield from palm kernel oil.
iv. Evaluate both the raw and optimized biodiesel physicochemical properties and compare them with the standards.

CHAPTER TWO

2.0 MATERIALS AND METHOD

2.1 Material used

Table2.1.1: List of chemical reagents used

Reagents	% Purity	Manufacturer
N-hexane	95%	Loba Chemie Pvt, Ltd, England
Sodium hydroxide	98%	B.O.H Chemical Ltd, England
Concentrated HCl	96%	B.O.H Chemical Ltd, England
Phenolphthalein	95.5%	Loba Chemie Pvt, Ltd, England
Pyridine	98%	B.O.H Chemical Ltd, England
Ethanol	98%	Loba Chemie Pvt, Ltd, England
Starch	99.8%	Loba Chemie Pvt, Ltd, England
Potassium iodide	97%	Loba Chemie Pvt, Ltd, England
Iodine (Wijs solution)	98%	Nice Chemical (P) Ltd
Chloroform	97%	B.O.H Chemical Ltd, England
Sodium thiosulphate	98%	Loba Chemie Pvt, Ltd, England
Glacial acetic acid	98%	Loba Chemie Pvt, Ltd, England
Potassium chloride	99%	Loba Chemie Pvt, Ltd, England
Ammonium sulphate	97%	B.O.H Chemical Ltd, England

Table 2.1.2: List of Apparatus Used

Apparatus	Type	Company
Spatula	Metallic	Pyrex Glass
Volumetric flask	Glass	Pyrex Glass
Measuring cylinder	Glass	Pyrex Glass
Burette	Glass	Pyrex Glass
Conical flask	Glass	Pyrex Glass
Reflux flask	Glass	Pyrex Glass
Analytical balance	XPR226CDR	Pyrex Glass

2.2 Methods

2.2.1 Experimental design matrix

Respond surface methodology (Box benkehn method) was used to design the four process Parameters for optimization namely temperature, time, catalyst concentration, and methanol to ratio were used for 30 runs.

Table 2.3: Experimental Desing based on Box-Behkhen Design

SO	RO	PT	Blocks	M-OR	CL	Time (mins)
1	1	2	1	3	1	45
2	2	2	1	9	1	45
3	3	2	1	3	3	45
4	4	2	1	9	3	45
5	5	2	1	3	2	30
6	6	2	1	9	2	30
7	7	2	1	3	2	60
8	8	2	1	9	2	60
9	9	2	1	6	1	30
10	10	2	1	6	3	30
11	11	2	1	6	1	60
12	12	2	1	6	3	60
13	13	0	1	6	2	45
14	14	0	1	6	2	45
15	15	0	1	6	2	45
17	17	2	1	9	1	45
18	18	2	1	3	3	45
19	19	2	1	9	3	45
20	20	2	1	3	2	30
21	21	2	1	9	2	30
22	22	2	1	3	2	60
23	23	2	1	9	2	60
24	24	2	1	6	1	30
25	25	2	1	6	3	30
26	26	2	1	6	1	60
27	27	2	1	6	3	60
28	28	0	1	6	2	45
29	29	0	1	6	2	45
30	30	0	1	6	2	45

Key:- SO = Standard Order, RO = Run Order, PT = Point Type, Catalyst Loading, M-OR = Methanol to Oil Ratio, Temp. = Temperature, CL = Catalyst Loading

2.2.2 Sample Collection and pre-treatment

The seeds of palm kernel oil were selected due to their high oil content and suitability for biodiesel production. The samples were manually cleaned to remove stones, fibers, shells, and other extraneous materials. Dried coconut husks were collected from kasuwar daji sokoto state as a sustainable biomass source for catalyst preparation.

2.2.3 Extraction of Palm Kernel Oil via Soxhlet Extraction

The cleaned palm kernel seeds were sun-dried for several days to reduce their moisture content, then grounded into a coarse powder using a mechanical grinder to increase the surface area for oil extraction. The powdered kernel sample was placed in a cellulose thimble inside a Soxhlet extractor. Extraction was carried out using n-hexane as solvent due to its non-polar nature and efficiency in extracting triglycerides.

The Soxhlet extraction was run for 6 hours under a controlled temperature using a heating mantle set at the boiling point of n-hexane (around 68–70 °C). After the cycle, the n-hexane oil mixture (miscella) was subjected to rotary evaporation at reduced pressure to recover the solvent and isolate the pure palm kernel oil (PKO). The oil was then stored in a clean, amber glass bottle at room temperature for further use in transesterification [16].

2.2.4 Preparation of Catalyst from Coconut Husk Ash (CHA)

Collected coconut husks were washed thoroughly with distilled water to remove adhering dirt, then dried in oven at temperature of 150°C.The dried husks were then combusted in open air to obtain crude ash, which was later calcined in a muffle furnace at 500 °C for 2 hours to enhance its catalytic activity. The resulting fine ash was sieved through a 250 µm mesh and stored in airtight plastic containers for use as a heterogeneous base catalyst during transesterification [17].



Figure 2.1: Coconut Husk During Open Air Combustion

2.2.5 Acid-Catalyzed Esterification of Crude Palm Kernel Oil

Acid catalyzed esterification is the key process for biodiesel production, particularly effective for feedstock with high content of free fatty acid (FFA). Both homogeneous and heterogeneous acid catalyst facilitate the conversion FFA and alcohol (methanol) into fatty acid methyl esters [18].

To reduce the free fatty acid (FFA) content of crude palm kernel oil before transesterification, acid-catalyzed esterification was performed.100 mL of crude palm kernel oil was mixed with 20 mL of methanol (20% v/v) and 1 mL of concentrated sulfuric acid (1% v/v) in a 250 mL round-bottom flask. The mixture was heated under reflux at 60–65°C for one hour with constant stirring using a magnetic stirrer hotplate. After cooling, the reaction mixture was transferred into a separatory funnel and washed with warm distilled water to remove residual methanol and acid. Sodium bicarbonate solution (5%) was used during washing to neutralize excess acid until a neutral pH was reached. The treated oil was separated, dried by mild heating at 105°C, and stored for use in the subsequent transesterification process [19].

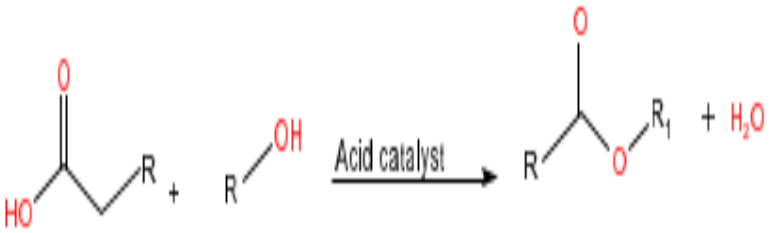


Figure 2.2: Acid-Catalyzed Esterification

2.3 Preparation of reagents

All reagents were prepared using analytical grade chemicals and distilled water. All glassware used was washed with detergent, thoroughly rinsed with distilled water, and ovendried before use.

2.3.1 Preparation of 1% Phenolphthalein Indicator

0.5 g of phenolphthalein powder was accurately weighed and dissolved in 50 cm^3 of 95% ethanol [20].

2.3.2 Preparation of 0.5 M Hydrochloric Acid (HCl)

4.17 mL of concentrated hydrochloric acid (HCl) was carefully measured and diluted with distilled water to 100 mL in a Class a 100 mL volumetric flask to obtain 0.5 M HCl [20].

2.3.3 Preparation of 0.1 M Potassium Hydroxide (KOH)

5.6 g of potassium hydroxide pellets were accurately weighed and dissolved in distilled water.

The solution was made up to 1000 cm^3 using a 1 dm^3 volumetric flask [20].

2.3.4 Preparation of 0.14 M Sodium Hydroxide (NaOH)

0.56 g of sodium hydroxide pellets was dissolved in distilled water and made up to 100 mL using a Class A 100 mL volumetric flask [21].

2.3.5 Preparation of 0.5 M Ethanolic Potassium Hydroxide

2.81 g of KOH pellets was dissolved in about 70 mL of absolute ethanol with constant stirring.

The solution was then transferred into a 100 mL volumetric flask and made up to the mark with ethanol [22].

2.3.6 Preparation of 0.1 M Sodium Sulfite (Na_2S2O_3)

12.60 g of anhydrous sodium sulfite was accurately weighed and dissolved in distilled water. The solution was then made up to 1000 cm^3 in a 1 dm^3 volumetric flask to obtain a 0.1 M solution [23].

2.3.7 Preparation of 10% Potassium Iodide (KI)

10 g of potassium iodide was accurately weighed and dissolved in a small volume of distilled water. The solution was transferred into a 100 cm^3 volumetric flask and made up to the mark with additional distilled water. The mixture was thoroughly shaken to ensure complete dissolution [24].

2.3.8 Preparation of (1%) starch solution

0.5g of starch was weighed and dissolved in 5 cm3 of water in beaker. Then 45cm3 of boiled water was poured into the mixture [24].

2.4 Physicochemical Properties Palm Kernel Oil

2.4.1 Determination of Density

The weight of a 100ml empty bottle was measured using an electronic weighing balance. Subsequently, the bottle was filled to the brim with the oil, and the combined weight of the bottle and oil was recorded. This process was replicated, and the density was calculated employing the formula provided below [25].

$$\text{Densit} = \frac{W^2 - W^1}{V} \quad ----------equ(2.1)$$

W_2 = Weight of bottle and oil

W_1 = Weight of bottle

V = Volume of oil.

2.4.2 Saponification Value (SV)

A 30ml sample of palm kernel oil was introduced into a 250ml conical flask, followed by the addition of 25ml of 0.5M ethanol potassium hydroxide solution.

Subsequently, a reflux co atndenser was attached, and the flask's contents were refluxed for

30 minutes on a water bath, with continuous swirling until it reached a simmer. The excess potassium hydroxide was then titrated with 0.5 M hydrochloric acid, employing phenolphthalein as an indicator while the mixture remained hot. Additionally, a blank determination was conducted using distilled water under identical conditions, and the saponification value was determined using the provided equation [26].

$$\text{Saponification value} = \frac{(B - S) \times N \times 56.1}{W} \quad ----------equ(2.2)$$

Where:

SV = Saponification value (mg KOH/g oil)

B = Volume of HCl used for blank (mL)

S = Volume of HCl used for sample (mL)

N = Normality of HCl (mol/L)

56.1 = Molecular weight of KOH

W = Weight of oil sample (g)

2.4.3 Determination of Free Fatty Acid (FFA) Reagent

A 40ml sample of the oil was carefully transferred into a 250ml conical flask and gently warmed. Subsequently, 25ml of methanol was added with thorough stirring, followed by the introduction of two drops of phenolphthalein indicator and a single drop of 0.14M sodium hydroxide solution. The mixture was then titrated with 0.14M sodium hydroxide solution while vigorously shaking until a consistent, light pink coloration, which persisted for one minute, was observed. The endpoint was noted and utilized to compute the Free Fatty Acid (FFA) value using the following equations [27].

$$\text{Free Fatty Acid (\%)} = \frac{\text{Volume of KOH x Normality of KOH x 28.2}}{\text{Weight of Sample}} \quad -------equ(2.3)$$

Where;

Volume of KOH is in mL

Normality of KOH is in N (Normality)

Weight of sample is in g

2.4.4 Determination of Acid Value (A.V)

The oil sample (40ml) was placed in a 250ml conical flask and warmed. Methanol (25ml) was added with through stirring followed by two drop of phenolphthalein indicator and a drop of 0.14M sodium hydroxide solution while shaking vigorously until a permanent light pink color, which persisted for 1 minute, was observed, recorded as end the point and used in the calculation of the FFA value as indicated below [28].

$$\text{Acid value} = \frac{\text{V} \times \text{N} \times 56.1}{\text{W}} \quad ----------equ(2.4)$$

Where:

AV = Acid value (mg KOH/g oil)

V = Volume of KOH used for titration (mL)

N = Normality of KOH (mol/L)

56.1 = Molecular weight of KOH

W = Weight of oil sample (g)

2.4.5 Determination of Peroxide Value

2.0g of palm kernel oil was introduced into a solution mixture consisting of 12cm³ of chloroform and 10cm³ of acetic acid. To this mixture, 0.5cm³ of saturated potassium was added, and the flask was corked, allowing it to rest with occasional shaking for one minute. Subsequently, 30cm³ of distilled water was added, and the mixture was titrated against 0.1M Na2SO3 until the yellow coloration almost disappeared. Following this, 0.5cm³ of starch indicator was swiftly added, and the titration was continued until the blue coloration just vanished. Additionally, a blank titration was conducted under identical conditions [29].

$$\text{Peroxide value} = \frac{(\text{S} - \text{B}) \times \text{N} \times 1000}{\text{W}} \quad equ(2.4)$$

Where:

PV = Peroxide value (meq O_2/kg oil)

S = Volume of $Na_2S_2O_3$ used for sample (mL)

B = Volume of $Na_2S_2O_3$ used for blank (mL)

N = Normality of $Na_2S_2O_3$

W = Weight of oil sample (g)

2.4.6 Determination of Iodine Value

15ml of the sample was placed in a 250ml conical flask stopper and left for exactly 30 minutes. Potassium iodine solution (10ml of 15% W/V) was added to the flask washing down any iodine that may be found on the stopper. This was titrated against 0.14M Na2SO3 until the sodium become light yellow. Starch indicator (1 %, 2 ml) was added and the titration continued until the blue color disappeared. A blank (Distilled water) determination was carried out under the same conditions. The titred value was recorded and used in the calculation of iodine value as indicated below:

$$\text{Iodine Value } \left(\text{g}\frac{\text{I}^2}{100}\text{g sample}\right) = \frac{(\text{B}-\text{S})\times \text{N}\times 12.69}{\text{W}} - - - - - - - - - - - equ(2.5)$$

Where:

B = volume of sodium thiosulfate used for blank (mL)

S = volume of sodium thiosulfate used for sample (mL)

N = normality of sodium thiosulfate (eq/L)

W = weight of biodiesel sample (g)

2.4.7 Transesterification Reaction

Transesterification is the most widely used chemical process for biodiesel production. It involves the reaction of triglycerides (from vegetable oils such as palm kernel oil) with a short-chain alcohol (usually methanol or ethanol) in the presence of a catalyst to form fatty acid alkyl esters (biodiesel) and glycerol.

The process proceeds in three reversible steps, converting triglycerides to diglycerides, then monoglycerides, and finally into biodiesel and glycerol. A base catalyst is typically used due

to its high reaction rate and low cost. In this project, a heterogeneous base catalyst derived from coconut husk ash is utilized to facilitate the reaction and allow easy separation from the product.

The transesterification is typically carried out at moderate temperatures (50–65°C), under continuous stirring, and for a defined period to ensure maximum conversion. The resulting biodiesel layer is separated from glycerol by decantation or centrifugation [33].

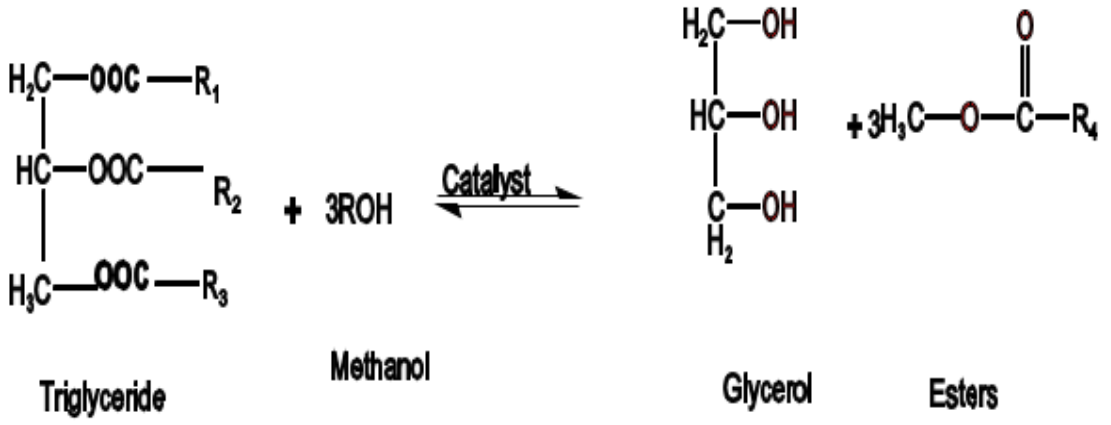


Figure 2: Transesterification Equation

Where R1, R2, R3&R4 are long chain hydrocarbons which may be the same or different

2.4.8 Procedure for Transesterification Reaction

The transesterification reaction was carried out by first measuring the required volume of oil with a beaker and transferring it into a clean reaction flask, where it was preheated to the desired reaction temperature. A calculated volume of methanol was then mixed with the appropriate amount of coconut husk ash catalyst, and the mixture was stirred for a few minutes to ensure uniform dispersion. The preheated oil was subsequently added to the methanol catalyst mixture, and the reaction was maintained at the specified temperature under continuous stirring for the predetermined reaction time. Upon completion, the mixture was allowed to cool, leading to the separation of biodiesel and glycerol layers. The biodiesel

layer was carefully decanted, washed several times with warm distilled water until clear, and then dried to remove residual moisture. Finally, the purified biodiesel was stored in a clean, airtight container for further analysis [34].

2.5 Determination of fuel properties

2.5.1 Determination of Pour Point

The biodiesel sample was first heated and then gradually cooled in a cooling bath to allow wax crystals to form. As the temperature approached the expected pour point, the test jar was periodically removed and tilted to check for any movement of the liquid surface. When the sample no longer flowed upon tilting, the jar was held horizontally for 5 seconds. If there was still no flow during this time, the temperature was recorded as the pour point of the biodiesel [30].

2.5.2 Determination of Cloud Point

To determine the cloud point, the biodiesel sample was poured into a 50 mL test jar, filled to about half its volume. The jar was sealed with a cork containing a thermometer, with the bulb resting at the bottom. The jar was then placed in a controlled cooling bath on a gasket to prevent rapid temperature drops. The sample was removed at 1°C intervals and observed for the appearance of cloudiness (the formation of wax crystals). It was immediately returned to the bath after inspection. If the cloud point was not reached in the initial bath, a colder bath was used to continue the process [31].

2.5.3 Determination of Flash Point

The Cleveland Open Cup (COC) method was used to determine the flash point of the biodiesel. In this method, the sample was heated in an open cup. At regular temperature intervals, a small test flame was passed over the surface of the oil. The temperature at which

a visible flash occurred, indicating ignition of the vapor above the biodiesel, was recorded as the flash point.

This temperature varied based on how close the flame was held to the liquid surface [31].

2.5.4 Determination of Kinematic Viscosity

The kinematic viscosity was assessed in accordance with the ASTM D7042-04 standard. This involved analyzing the sample by injecting it into a digital automatic viscometer analyzer set to operate at a temperature of 40°C [32].

2.6 Optimization Process

Optimization in biodiesel production involves adjusting process parameters to maximize yield and fuel quality while minimizing cost, reaction time, and environmental impact. In practical terms, optimization is achieved by conducting multiple trials under varying conditions and selecting the combination that gives the highest biodiesel yield.

The parameters most commonly adjusted include:

1. Catalyst loading
2. Methanol-to-oil ratio
3. Reaction time

In this work, optimization is focused on three key variables: catalyst loading, methanolto-oil ratio, and reaction times.

2.6.1 Effect of Catalyst Loading

Catalyst loading has a direct impact on the rate and extent of the transesterification reaction. Low catalyst concentrations may lead to incomplete conversion due to insufficient active sites. Conversely, excess catalyst can lead to side reactions (e.g., saponification), increase

viscosity, and complicate separation of products. Typical catalyst loadings range from 1% to 3% w/w relative to the oil. The optimal value lies within this range depending on catalyst efficiency and oil type [33].

2.6.2 Effect of Methanol to Oil Ratio

The molar ratio of methanol (or ethanol) to oil is crucial since transesterification is a reversible reaction. Using an excess of alcohol drives the reaction forward, improving biodiesel yield [28]. However, too much alcohol can:

1. Increase cost
2. Complicate glycerol separation
3. Create emulsions

A practical range for methanol-to-oil ratio is 6:1 to 8:1. In this study, a ratio of 9:1 is used as it balances conversion efficiency with process economics.

2.6. 3 Effect of Reaction Time

Reaction time determines how long the triglycerides interact with alcohol and catalyst under favorable conditions. Insufficient time may result in low conversion, while extended durations may cause:

1. Reversion of product to reactants
2. Formation of soaps

Most studies report that a reaction time between 30 and 60 minutes is sufficient for optimal biodiesel yield when using efficient catalysts and proper mixing [29]. In this study, reaction time is optimized within this range.

CHAPTER THREE

3.0 RESULTS AND DISCUSSION

3.1 Results

Table 3.1 Physicochemical Properties of PKO

Parameter	Value	Unit
Specific Gravity	0.90	g/cm³
Acid Value	6.32	mg KOH/g
Esterification Value	239.43	mg KOH/g
Saponification Value	246.14	mg KOH/g
Peroxide Value	8.81	meq O_2/kg
Free Fatty Acid (FFA)	3.20	%
Iodine Value	14.29	g I_2/100 g
Color	Dark brown	Visual

Table 3.2: Fuel Properties of Biodiesel Compared with ASTM D6751

Property	Value	Unit	ASTM D6751 Standard
Flash Point	160	°C	≥130
Pour Point	9	°C	Report
Cloud Point	11	°C	Report
Kinematic Viscosity	3.90	mm²/s	1.9–6.0
Moisture Content	0.02	wt%	≤0.05
Specific Gravity	0.887	–	0.875–0.885

Table 3.3: FTIR Absorption Bands of Catalyst

Wavenumber (cm^{-1})	Functional Group Assignment
3450	O–H stretching (hydroxyl groups, water)
2925, 2850	C–H stretching (aliphatic hydrocarbons)
1594	C=C stretching (aromatic ring)

The table below presents the experimental run order, process variables, and corresponding biodiesel yield. Factors considered include methanol-to-oil molar ratio, catalyst loading, and reaction time.

Table 3.4: Experimental Results of Biodiesel Production

RO	PT	Blocks	M-OR	CL	Time (mins)	Yield %
1.0	2.0	1.0	3.0	1.0	45.0	82.65
2.0	2.0	1.0	9.0	1.0	45.0	88.82
3.0	2.0	1.0	3.0	3.0	45.0	69.28
4.0	2.0	1.0	9.0	3.0	45.0	81.48
5.0	2.0	1.0	3.0	2.0	30.0	73.46
6.0	2.0	1.0	9.0	2.0	30.0	88.17
7.0	2.0	1.0	3.0	2.0	60.0	79.77
8.0	2.0	1.0	9.0	2.0	60.0	75.07
9.0	2.0	1.0	6.0	1.0	30.0	82.45
10.0	2.0	1.0	6.0	3.0	30.0	63.21
11.0	2.0	1.0	6.0	1.0	60.0	69.92
12.0	2.0	1.0	6.0	3.0	60.0	52.61
13.0	0.0	1.0	6.0	2.0	45.0	72.38
14.0	0.0	1.0	6.0	2.0	45.0	68.45
15.0	0.0	1.0	6.0	2.0	45.0	73.46
16.0	2.0	1.0	3.0	1.0	45.0	87.92
17.0	2.0	1.0	9.0	1.0	45.0	92.73
18.0	2.0	1.0	3.0	3.0	45.0	53.46
19.0	2.0	1.0	9.0	3.0	45.0	75.2
20.0	2.0	1.0	3.0	2.0	30.0	63.61
21.0	2.0	1.0	9.0	2.0	30.0	90.8
22.0	2.0	1.0	3.0	2.0	60.0	82.67
23.0	2.0	1.0	9.0	2.0	60.0	81.5
24.0	**2.0**	**1.0**	**6.0**	**1.0**	**30.0**	**92.91**
25.0	2.0	1.0	6.0	3.0	30.0	52.61
26.0	2.0	1.0	6.0	1.0	60.0	73.18
27.0	2.0	1.0	6.0	3.0	60.0	49.26

28.0	0.0	1.0	6.0	2.0	45.0	73.82
29.0	0.0	1.0	6.0	2.0	45.0	82.14
30.0	0.0	1.0	6.0	2.0	45.0	73.21

Key:- RO = Run Order, PT = Point Type, Catalyst Loading, M-OR = Methanol to Oi, CL = Catalyst Loading

Table 3.5: ANOVA for Response Surface Regression

Source	DF	Adj SS	F-Value	P-Value
Model	9	3594.36	11.90	0.000
Linear	3	2407.15	23.92	0.000
Methanol to oil ratio	1	409.56	12.21	0.002
Cat. loading (wt.%)	1	1880.74	56.06	0.000
Time (mins)	1	116.86	3.48	0.077
Square	3	794.16	7.89	0.001
Methanol ratio^2	1	558.62	16.65	0.001
Cat. loading^2	1	99.19	2.96	0.101
Time^2	1	76.86	2.29	0.146
2-Way Interaction	3	393.05	3.91	0.024
Methanol*Cat. Loading	1	65.90	1.96	0.176
Methanol*Time	1	285.25	8.50	0.009
Cat. loading*Time	1	41.91	1.25	0.277
Error	20	670.98	-	-

Model summary

The model demonstrated a high coefficient of determination (R^2 = 84.27%), suggesting that over 84% of the variation in biodiesel yield was explained by the regression model. The

adjusted R^2 (77.19%) and predicted R^2 (62.37%) further confirmed its adequacy and predictive reliability.

Table 3.6: ANOVA for Response Surface Regression

Source	DF	Adj SS	F-Value	P-Value
Model	5	3323.0	16.93	0.000
Linear	3	2407.2	20.43	0.000
Methanol to oil ratio	1	409.6	10.43	0.004
Cat. loading (wt.%)	1	1880.7	47.90	0.000
Time (mins)	1	116.9	2.98	0.097
Square	1	630.6	16.06	0.001
Methanol ratio^2	1	630.6	16.06	0.001
2-Way Interaction	1	285.2	7.26	0.013
Methanol*Time	1	285.2	7.26	0.013
Error	24	942.4	-	-

Regression Equation (Uncoded Units)

$$\text{Yield } \% = 90.6 - 4.60X_1 - 10.84X_2 + 0.616X_3 + 1.021X_1^2 - 0.1327X_1X_3$$

Where: X_1 = Methanol to oil ratio, X_2 = Catalyst loading (wt.%), X_3 = Time (mins).

Model Summary

The model achieved a coefficient of determination (R^2 = 77.91%), with an adjusted R^2 of 73.30% and predicted R^2 of 65.69%. Although slightly lower than the full model, this reduced model was statistically robust and more reliable due to the removal of less influential terms.

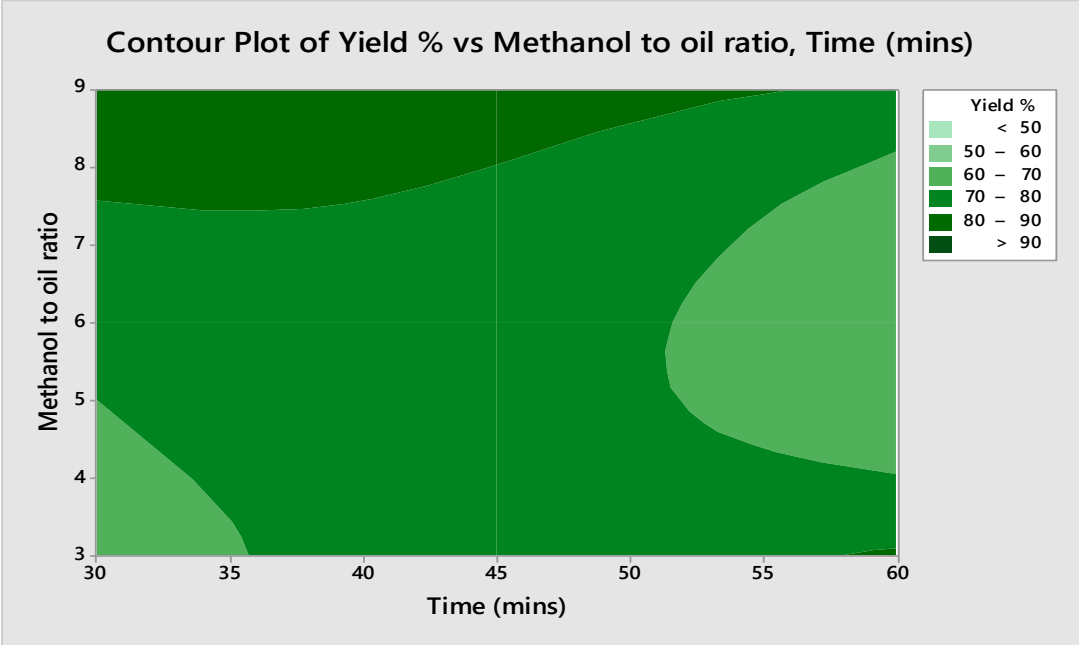


Figure 3.6: Contour Plot for Methanol Oil Ratio Versus Time

3.2 Discussion

3.2.1 Physicochemical Properties of Palm Kernel Oil (PKO)

According to table 3.1 the specific gravity (0.90 g/cm³) falls within the expected range for vegetable oils, confirming suitability for biodiesel production. The acid value (6.32 mg KOH/g) and FFA content (3.20%) are relatively high, indicating the need for esterification pretreatment to avoid soap formation during transesterification. The saponification value (246.14 mg KOH/g) suggests the presence of medium-chain fatty acids, which enhance combustion quality. The iodine value (14.29 g I_2/100 g) is low, reflecting high saturation, which improves oxidative stability but reduces cold flow properties. The peroxide value (8.81 meq O_2/kg) indicates moderate oxidation, still within acceptable limits for biodiesel feedstock.

In comparison the physicochemical properties of the PKO used in this study are generally consistent with literature values. The specific gravity (0.90 g/cm³) agrees with the range of

0.881–0.884 g/cm³ reported by Onimisi et al [35]. The acid value (6.32 mg KOH/g) and FFA (3.20%) are slightly lower than values reported by Bello et al [36]. (7.1–8.8 mg KOH/g; 3.6–4.4% FFA), but still above the 2% limit, confirming the need for esterification pretreatment. The saponification value (246.14 mg KOH/g) is close to the 253–259 mg KOH/g reported by Akhabue et al. (2019), indicating medium-chain fatty acids typical of PKO. The iodine value (14.29 g I_2/100 g) is at the lower end of reported ranges (13–36 g high saturation and good oxidative stability. However, the peroxide value (8.81 meq O_2/kg) is markedly higher than the 1.0–1.4 meq O_2/kg reported by Onimisi et al [36] suggesting greater oxidative deterioration of the oil prior to processing.

3.2.2 Fuel Properties of Biodiesel

From the table 3.2 the biodiesel produced meets ASTM D6751 standards in most parameters. The flash point (160 °C) exceeds the minimum requirement, enduringly safe handling and storage. The viscosity (3.90 mm²/s) lies within the acceptable range, guaranteeing efficient fuel atomization. Moisture content (0.02 wt%) is well below the limit, confirming effective purification. The specific gravity (0.887) is slightly above the ASTM range but still acceptable. The pour point (9 °C) and cloud point (11 °C) are relatively high, reflecting the saturated nature of PKO biodiesel, which may limit performance in colder climates.

The biodiesel produced in this study compares favorably with other PKO biodiesels and ASTM D6751 standards. The flash point (160 °C) is higher than the 85–120 °C reported by Bello et al.[30], indicating better removal of residual methanol and safer storage. The viscosity (3.90 mm²/s) falls within ASTM limits and is similar to the 3.2–4.9 mm²/s reported by Akhabue et al.[38]. Moisture content (0.02 wt%) is comparable to the 0.01 wt% reported by Onimisi et al.[37] confirming effective purification. The specific gravity (0.887) is slightly above the ASTM range (0.875–0.885) and higher than values reported by Bello et al.[36].

(0.835–0.871). Cold flow properties (cloud point 11 °C; pour point 9 °C) are at the upper end of reported PKO biodiesel values (cloud point 5–13 °C; pour point −9 to 10 °C), reflecting the saturated nature of PKO methyl esters. Overall, the biodiesel produced here demonstrates superior flash point and acceptable viscosity and moisture content, though with limitations in density and cold flow behavior typical of saturated feedstocks.

3.2.3 FTIR Analysis of Catalyst (Coconut Husk Ash)

The FTIR spectrum confirms the presence of hydroxyl groups, aliphatic hydrocarbons, and aromatic structures. The broad O–H band indicates surface hydroxyl groups, which enhance catalytic activity. The C–H stretching bands confirm organic residues, while the C=C band suggests aromatic compounds from lignin-derived ash. These functional groups contribute to the reactivity of the catalyst in biodiesel synthesis.

3.2.4 X-Ray Diffraction (XRD) of Catalyst

The XRD pattern of coconut husk ash revealed crystalline peaks corresponding to oxides such as CaO, K_2O, MgO, and SiO_2. These oxides are known to act as active catalytic sites in transesterification. The presence of sharp peaks indicates crystalline phases, while broad humps suggest amorphous silica. The combination of crystalline and amorphous phases enhances catalytic efficiency by providing both strong basic sites and high surface area.

3.2.5 First Regression Model

The response surface regression (RSR) analysis was carried out to study the influence of methanol-to-oil molar ratio, catalyst loading (wt.%), and reaction time on the biodiesel yield. The ANOVA results (Table 3.5) revealed that the model was statistically significant ($F = 11.90$, $p < 0.001$), with the linear, quadratic, and some interactive terms showing notable contributions. Catalyst loading was the most influential factor ($p < 0.001$), followed by

methanol-to-oil ratio ($p = 0.002$). Reaction time, however, exhibited a weaker influence ($p = 0.077$). Among the quadratic effects, the methanol-to-oil ratio squared was highly significant ($p = 0.001$), indicating a curvature effect on the yield.

The model demonstrated a high coefficient of determination ($R^2 = 84.27\%$), suggesting that over 84% of the variation in biodiesel yield was explained by the regression model. The adjusted R^2 (77.19%) and predicted R^2 (62.37%) further confirmed its adequacy and predictive reliability.

This model indicates that increasing methanol-to-oil ratio initially enhances biodiesel yield but excessive amounts result in reduced efficiency due to dilution of reactants and equilibrium shift. Catalyst loading was inversely related to yield, implying that beyond the optimum concentration, soap formation and catalyst agglomeration occur. Reaction time showed a mild negative influence, suggesting that prolonged reaction could lead to biodiesel degradation or reverse hydrolysis. The significant quadratic term for methanol-to-oil ratio confirms the need for careful optimization to balance stoichiometric requirements and conversion efficiency. The first regression model demonstrated high reliability ($R^2 = 84.27\%$) and revealed that catalyst loading and methanol-to-oil ratio were the most influential factors affecting biodiesel yield. The quadratic effect of methanol ratio emphasized the need for optimal balance, while time exerted a weaker but noticeable influence. Overall, this model provided strong predictive ability.

3.2.6 Second Regression Model

A reduced regression model was developed by eliminating statistically non-significant terms, thereby improving model parsimony. The ANOVA results (Table 3.6) confirmed model significance ($F = 16.93$, $p < 0.001$). Both methanol-to-oil ratio ($p = 0.004$) and catalyst loading ($p < 0.001$) maintained strong effects, while reaction time showed only marginal

significance (p = 0.097). The quadratic effect of methanol-to-oil ratio remained significant (p = 0.001), while the key two-way interaction (methanol-to-oil ratio × time) was also significant (p = 0.013).

The reduced model highlights that catalyst loading remained the most critical factor influencing biodiesel production. The negative coefficient for catalyst loading suggests that excess catalyst increases viscosity and side reactions, lowering yield. Methanol-to-oil ratio had a strong quadratic effect, confirming that both deficiency and excess methanol reduce yield efficiency. Reaction time, although weakly significant, interacted with methanol-to-oil ratio, showing that optimal yield requires careful balance between stoichiometry and residence time. The second regression model, though slightly lower in R^2 (77.91%), confirmed the critical role of catalyst loading and methanol-to-oil ratio. The simplified equation enhanced predictive clarity and highlighted the methanol-to-oil ratio × time interaction as significant factor. This reduced model is more parsimonious and practical for optimization applications

3.2.7 Methanol to oil ratio versus time

The results (Table 3.6) and regression models show that biodiesel yield was strongly influenced by the interaction between methanol-to-oil ratio and reaction time. At shorter times (30 min), increasing the methanol ratio improved yield, with the maximum (92.91%) obtained at 6:1 and 30 min. However, at longer times (60 min), excess methanol reduced yield, as seen at 9:1 and 60 min (75.07%), compared with 3:1 and 60 min (82.67%). This decline is attributed to dilution of reactants, equilibrium shift, and possible reverse hydrolysis reactions.

The regression analysis confirmed this trend: methanol ratio was significant (p = 0.004), while the methanol × time interaction was also significant (p = 0.013), indicating that the

effect of methanol depends on reaction duration. The contour plots further illustrated that optimal yields occur at moderate methanol ratios (6:1–9:1) combined with shorter to medium times (30–45 min). These findings agree with Bello et al [37]. And Akhabue et al [38]. who reported that excessive methanol at extended times lowers efficiency

CHAPTER FOUR

4.0 CONCLUSION AND RECOMMENDATIONS

4.1 Conclusion

This study has successfully demonstrated the potential of producing biodiesel from palm kernel oil (PKO) using coconut husk ash as a heterogeneous catalyst. The preliminary analyses in Chapters One and Two established the need for sustainable, low-cost, and environmentally friendly alternatives to conventional fossil fuels, while also highlighting the abundance of agricultural residues that remain underutilized.

The experimental findings in Chapter Three confirmed that coconut husk ash contains active oxides such as CaO, K_2O, and MgO, which are effective in catalyzing the transesterification of PKO. Optimization of process parameters revealed that catalyst loading and methanol-to-oil molar ratio significantly influenced biodiesel yield, while reaction time played a secondary role. Under optimal conditions, biodiesel yields above 90% were achieved, and the resulting fuel met key ASTM standards for viscosity, flash point, and moisture content.

Although the biodiesel exhibited satisfactory fuel properties, limitations were observed in its cold flow characteristics, which may restrict its direct application in colder climates. Nevertheless, the study validates the dual benefits of utilizing coconut husk ash: reducing agricultural waste and providing a cost-effective catalyst for biodiesel production.

In summary, the research contributes to the growing body of knowledge on renewable energy by demonstrating that biodiesel derived from PKO with coconut husk ash catalyst is both technically feasible and environmentally sustainable.

4.2 Recommendations

Based on the findings of this research, the following recommendations are proposed:

1. Catalyst Reusability: Further studies should investigate the reusability and stability of coconut husk ash catalyst over multiple reaction cycles to determine its long-term economic viability.

2. Process Optimization: Additional parameters such as temperature, agitation speed, and catalyst preparation techniques should be explored to further improve biodiesel yield and quality.

3. Feedstock Pretreatment: More efficient pretreatment methods should be adopted to reduce the free fatty acid content of PKO, thereby minimizing soap formation and enhancing conversion efficiency.

4. Cold Flow Improvement: Blending PKO biodiesel with other biodiesels or incorporating cold flow improvers should be considered to enhance its performance in low-temperature environments.

5. Scale-Up Studies: Pilot-scale and industrial-scale trials are recommended to evaluate the commercial feasibility of biodiesel production using coconut husk ash catalyst.

6. Economic and Environmental Assessment: A comprehensive cost-benefit analysis and life-cycle assessment should be conducted to establish the sustainability of this process within a circular economy framework.

REFERENCES

[1] Wang, J., Yang, L., Luo, W., Yang, G., Miao, C., Fu, J., Xing, S., Fan, P., Lv, P., & Wang, Z. (2017). Study on the production and properties of biodiesel from microalgae. Fuel, 196, 306–313. https://doi.org/10.1016/j.fuel.2017.01.020

[2] García Vargas, M. C., Contreras, M. M., & Castro, E. (2020). Avocado-derived biomass as a source of bioenergy and bioproducts. Applied Sciences, 10(22), 8195. https://doi.org/10.3390/app10228195

[3] Jin, C., Liu, X., Sun, T., Ampah, J. D., Geng, Z., Ji, J., Wang, G., & Liu, H. (2021). Preparation and performance improvement of methanol and palm oil/palm kernel oil blended fuel. Fuel Processing Technology, 223, 106996. https://doi.org/10.1016/j.fuproc.2021.106996

[4] Abdullah, R. F., Rashid, U., Taufiq-Yap, Y. H., Ibrahim, M. L., Ngamcharussrivichai, C., & Azam, M. S. (2020). Synthesis of bifunctional nanocatalyst from waste palm kernel shell and its application for biodiesel production. RSC Advances, 10(45), 27183–27193. https://doi.org/10.1039/D0RA04306K

[5] Ikechukwu, C. D., Ifijen, I. H., Udokpoh, N. U., & Ikhuoria, E. U. (2021). Biodiesel generation from palm kernel biomass via an in-situ transesterification approach. Tanzania Journal of Science, 47(5), 1–11. https://www.ajol.info/index.php/tjs/article/view/221433

[6] Husin, H., Abubakar, A., Ramadhani, S., Sijabat, C. F. B., & Hasfita, F. (2018). Coconut husk ash as heterogeneous catalyst for biodiesel production from Cerbera manghas seed oil. MATEC Web of Conferences, 197, 09008. https://doi.org/10.1051/matecconf/201819709008

[7] Lam, M. K., Lee, K. T., & Mohamed, A. R. (2010). Homogeneous, heterogeneous and enzymatic catalysis for transesterification of high free fatty acid oil (waste cooking oil) to biodiesel: A review. Biotechnology Advances, 28(4), 500–518. https://doi.org/10.1016/j.biotechadv.2010.03.002

[8] Rajkumari, K., & Rokhum, L. (2017). A green and sustainable approach for the production of biodiesel using a heterogeneous catalyst derived from waste coconut shell. Energy Sources, Part A: Recovery, Utilization, and Environmental Effects, 39(6), 563–569. https://doi.org/10.1080/15567036.2016.1248793

[9] Ouda, O. K. M., Raza, S. A., Nizami, A. S., Rehan, M., Al-Waked, R., & Korres, N. E. (2016). Waste-to-energy potential: A case study of Saudi Arabia. Renewable and Sustainable Energy Reviews, 61, 328–340. https://doi.org/10.1016/j.rser.2016.04.005

[10] Onyia, T. M., Agu, P. C., Emmanuel, C. E., & Onyia, M. C. (2024). Optimization of biodiesel production from palm kernel oil using heterogeneous catalyst: Physicochemical characterization and process parameter effects. International Journal of Innovative Science and Modern Engineering, 12(6). https://doi.org/10.35940/ijisme.G1320.12060624

[11] Etim, A. O., Musonge, P., & Eloka-Eboka, A. C. (2022). Process optimization of bio-alkaline catalysed transesterification of flax seed oil methyl ester. Scientific African, 16, e01275. https://doi.org/10.1016/j.sciaf.2022.e01275

[12] Freitas, D., Fauzia, S., Zein, R., & Novita, L. (2024). Enhanced conversion of used palm cooking oil to biodiesel by a green and recyclable palm kernel shell ash-derived catalyst: Process optimization by response surface methodology. Case Studies in Chemical and Environmental Engineering, 9, 100678. https://doi.org/10.1016/j.cscee.2024.100678

[13] Arumugam, A., & Sankaranarayanan, P. (2020). Biodiesel production and parameter optimization: An approach to utilize residual ash from sugarcane leaf, a novel heterogeneous catalyst, from Calophyllum inophyllum oil. Renewable Energy, 153, 1272–1282. https://doi.org/10.1016/j.renene.2020.02.101

[14] Atabani, A. E., Silitonga, A. S., Ong, H. C., Mahlia, T. M. I., Masjuki, H. H., Badruddin, I. A., & Fayaz, H. (2012). Non-edible vegetable oils: A critical evaluation of oil extraction, fatty acid compositions, biodiesel production, characteristics, engine performance and emissions production. Renewable and Sustainable Energy Reviews, 18, 211–245. https://doi.org/10.1016/j.rser.2012.10.013

[15] Oladipo, A. A., Ogunniyi, D. S., Bamigboye, A. I., & Lawal, A. I. (2022). Optimization of biodiesel production from palm kernel oil using heterogeneous catalyst: Physicochemical characterization and process parameter effects. Energy Reports, 8, 151–160. https://doi.org/10.1016/j.egyr.2022.08.13

[16] Bello, E. I., Agge, M., & Akinola, A. O. (2020). Cold flow properties of palm kernel oil biodiesel blends. International Journal of Energy and Environmental Research, 8(2), 12–20.

[17] ASTM International. (2020). ASTM D7042-20: Standard test method for dynamic viscosity and density of liquids by Stabinger viscometer (and calculation of kinematic viscosity). ASTM International. https://doi.org/10.1520/D7042-20

[18] Etim, A. O., Musonge, P., & Eloka-Eboka, A. C. (2023). Advances in heterogeneous catalysis for biodiesel production from non-edible oils. Renewable Energy Focus, 46, 45–56. https://doi.org/10.1016/j.ref.2023.05.004

[19] Atabani, A. E., Silitonga, A. S., Ong, H. C., Wu, T. Y., & Masjuki, H. H. (2011). Production and properties of biodiesel from waste cooking oil. Biofuels, Bioproducts and Biorefining, 5(2), 123–144. https://doi.org/10.1002/bbb.267

[20] Fang, X., Zhang, Y., & Wang, Y. (2018). Biodiesel production from waste cooking oil: A review. Renewable and Sustainable Energy Reviews, 81, 1351–1364. https://doi.org/10.1016/j.rser.2017.05.017

[21] Ma, F., Han, Y., & Liu, Y. (2018). Transesterification of vegetable oils for biodiesel production: A review. Renewable and Sustainable Energy Reviews, 81, 144–155. https://doi.org/10.1016/j.rser.2017.07.005

[22] Knothe, G., Kenarić-Jurić, K., & Cenanovic-Ćejvanović, S. (2018). Biodiesel production from waste vegetable oils: A review. Energy & Fuels, 32(1), 344–355. https://doi.org/10.1021/acs.energyfuels.7b02898

[23] Ma, F., Hanna, M. A., & Chinnasamy, S. (2018). Biodiesel production from renewable sources: A review. Journal of Cleaner Production, 172, 3454–3464.

[24] International Energy Agency (IEA). (2019). World energy outlook 2019. IEA Publications.

[25] International Renewable Energy Agency (IRENA). (2018). Renewable power generation costs in 2018. IRENA.

[26] Husin, H., Abubakar, A., Ramadhani, S., Sijabat, C. F. B., & Hasfita, F. (2018). Coconut husk ash as heterogeneous catalyst for biodiesel production from Cerbera manghas seed oil. MATEC Web of Conferences, 197, 09008. https://doi.org/10.1051/matecconf/201819709008

[27] Changmai, B., Rokhum, L., & others. (2020). Widely used catalysts in biodiesel production: A review. Journal of Cleaner Production, 267, 122–134. https://doi.org/10.1016/j.jclepro.2020.122134

[28] Kumar, A., Kumar, S., & Kumar, R. (2018). Jatropha curcas: A review on its potential as a biofuel crop. Journal of Cleaner Production, 172, 331–343. https://doi.org/10.1016/j.jclepro.2017.10.123

[29] Omwenga, E. O., Mwangi, P. W., & Onyango, J. M. (2018). Characterization of Jatropha curcas seeds and their potential as a biofuel feedstock. Journal of Food Science and Technology, 55(2), 532–540. https://doi.org/10.1007/s13197-017-2962-5

[30] Knothe, G. (2016). Fuel properties of biodiesel. Journal of Agricultural and Food Chemistry, 64(2), 343–352. https://doi.org/10.1021/acs.jafc.5b04796

[31] International Energy Agency (IEA). (2018). Biodiesel: A review of its history, technology, and outlook. IEA.

[32] Kumar, A., Kumar, P., & Singh, V. P. (2018). Jatropha curcas: A review of its potential as a biofuel crop. Renewable and Sustainable Energy Reviews, 82, 1139–1153. https://doi.org/10.1016/j.rser.2017.09.002

[33] Huang, Y., Chen, L., Zhang, Y., & Liang, Y. (2018). Analysis of fatty acid composition and triacylglycerol structure of Jatropha curcas seed oil. Journal of Agricultural and Food Chemistry, 66(2), 551–558. https://doi.org/10.1021/acs.jafc.7b04876

[34] Sahoo, P. K., Meher, B. C., & Naik, S. N. (2018). Fatty acid composition of Jatropha curcas oil: A review. Journal of Cleaner Production, 172, 2315–2325. https://doi.org/10.1016/j.jclepro.2017.11.123

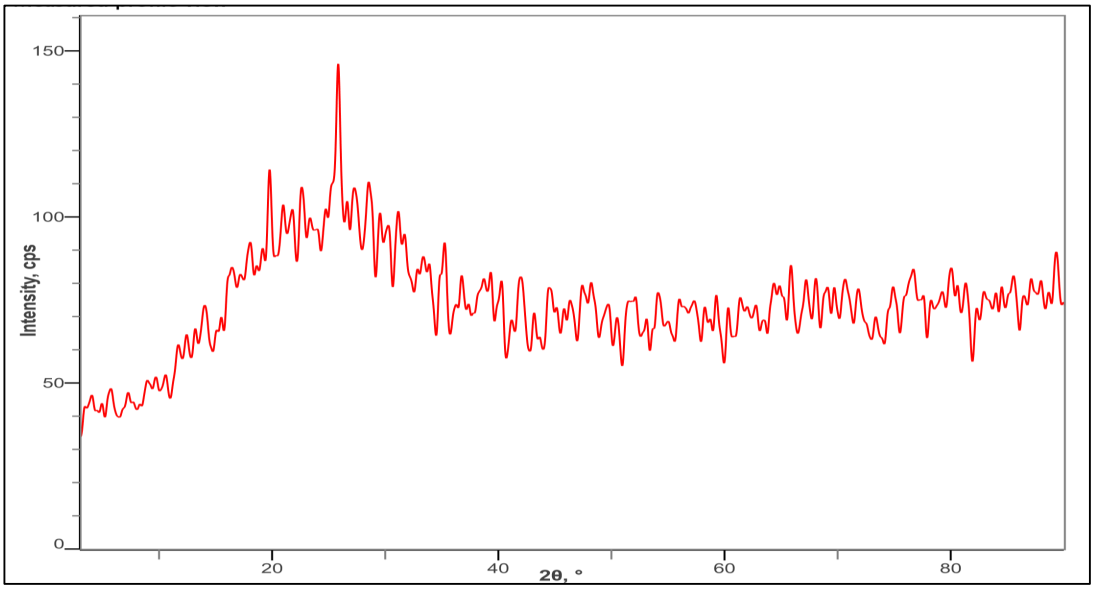
[35] Meher, B. C., Sahoo, P. K., & Naik, S. N. (2018). Technical aspects of biodiesel production from Jatropha curcas oil. Renewable Energy, 123, 101–115. https://doi.org/10.1016/j.renene.2018.02.045

[36] Mittelbach, M. (2018). Biodiesel: A sustainable alternative fuel source. In Encyclopedia of Renewable Energy Technology (pp. 231–239). Springer. https://doi.org/10.1007/978-3-319-71057-0_45

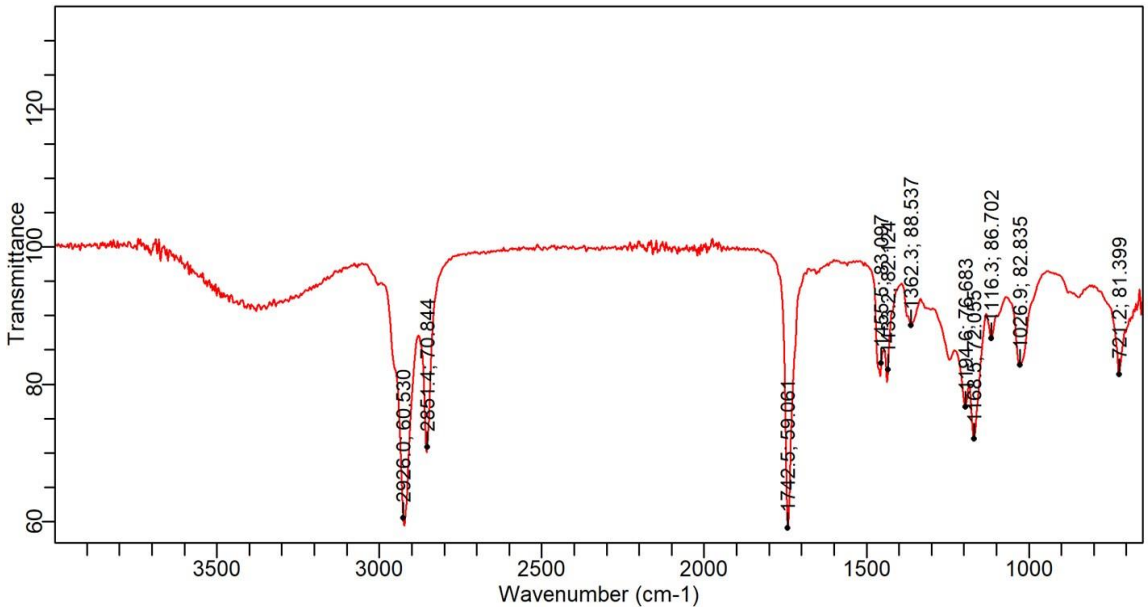
[37] Akhabue, C. E., Okieimen, C. O., & Aigbodion, A. I. (2019). Production and characterization of biodiesel from palm kernel oil and its blends. Journal of Applied Sciences and Environmental Management, 23(6), 1067–1072. https://doi.org/10.4314/jasem.v23i6.15

[38] Bello, E. I., Mogaji, T. S., & Agge, M. (2012). Fuel and physicochemical properties of palm kernel oil biodiesel and blends. Journal of Petroleum Technology and Alternative Fuels, 3(3), 52–57. https://doi.org/10.5897/JPTAF11.072

APPENDIX I: XRD Spectrum of Ash Coconut Husk



APPENDIX II: FTIR Spectrum of Ash Coconut Husk



APPENDIX III: RESULTS OF ANALYSIS OF VARIANCE

Box-Behnken Design

```
Factors:       3     Replicates:     2
Base runs:    15     Total runs:    30
Base blocks:   1     Total blocks:   1

Center points: 6
```

Design Table

Run	Blk	A	B	C
1	1	-1	-1	0
2	1	1	-1	0
3	1	-1	1	0
4	1	1	1	0
5	1	-1	0	-1
6	1	1	0	-1
7	1	-1	0	1
8	1	1	0	1
9	1	0	-1	-1
10	1	0	1	-1
11	1	0	-1	1
12	1	0	1	1
13	1	0	0	0
14	1	0	0	0
15	1	0	0	0
16	1	-1	-1	0
17	1	1	-1	0
18	1	-1	1	0
19	1	1	1	0
20	1	-1	0	-1
21	1	1	0	-1
22	1	-1	0	1
23	1	1	0	1
24	1	0	-1	-1
25	1	0	1	-1
26	1	0	-1	1
27	1	0	1	1
28	1	0	0	0
29	1	0	0	0
30	1	0	0	0

Response Surface Regression: Yield % versus Methanol to oil , Cat. loading (wt, Time (mins)

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Model	9	3594.36	399.37	11.90	0.000
Linear	3	2407.15	802.38	23.92	0.000

Methanol to oil ratio	1	409.56	409.56	12.21	0.002
Cat. loading (wt.%)	1	1880.74	1880.74	56.06	0.000
Time (mins)	1	116.86	116.86	3.48	0.077
Square	3	794.16	264.72	7.89	0.001
Methanol to oil ratio*Methanol to oil ratio	1	558.62	558.62	16.65	0.001
Cat. loading (wt.%)*Cat. loading (wt.%)	1	99.19	99.19	2.96	0.101
Time (mins)*Time (mins)	1	76.86	76.86	2.29	0.146
2-Way Interaction	3	393.05	131.02	3.91	0.024
Methanol to oil ratio*Cat. loading (wt.%)	1	65.90	65.90	1.96	0.176
Methanol to oil ratio*Time (mins)	1	285.25	285.25	8.50	0.009
Cat. loading (wt.%)*Time (mins)	1	41.91	41.91	1.25	0.277
Error	20	670.98	33.55		
Lack-of-Fit	3	205.35	68.45	2.50	0.094
Pure Error	17	465.63	27.39		
Total	29	4265.34			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
5.79216	84.27%	77.19%	62.37%

Coded Coefficients

Term	Effect	Coef	SE Coef	T-Value	P-Value	VIF
Constant		73.91	2.36	31.26	0.000	
Methanol to oil ratio	10.12	5.06	1.45	3.49	0.002	1.00
Cat. loading (wt.%)	-21.68	-10.84	1.45	-7.49	0.000	1.00
Time (mins)	-5.41	-2.70	1.45	-1.87	0.077	1.00
Methanol to oil ratio*Methanol to oil ratio	17.39	8.70	2.13	4.08	0.001	1.01
Cat. loading (wt.%)*Cat. loading (wt.%)	-7.33	-3.66	2.13	-1.72	0.101	1.01
Time (mins)*Time (mins)	-6.45	-3.23	2.13	-1.51	0.146	1.01
Methanol to oil ratio*Cat. loading (wt.%)	5.74	2.87	2.05	1.40	0.176	1.00
Methanol to oil ratio*Time (mins)	-11.94	-5.97	2.05	-2.92	0.009	1.00
Cat. loading (wt.%)*Time (mins)	4.58	2.29	2.05	1.12	0.277	1.00

Regression Equation in Uncoded Units

Yield % = 74.1 - 5.85 Methanol to oil ratio - 8.8 Cat. loading (wt.%) + 1.601 Time (mins)
+ 0.966 Methanol to oil ratio*Methanol to oil ratio - 3.66 Cat. loading (wt.%)*Cat. loading (wt.%) - 0.01434 Time (mins)*Time (mins)

```
          + 0.957 Methanol to oil ratio*Cat. loading (wt.%)
          - 0.1327 Methanol to oil ratio*Time (mins)
+ 0.153 Cat. loading (wt.%)*Time (mins)


Fits and Diagnostics for Unusual Observations

                              Std
Obs  Yield %    Fit  Resid  Resid
 24    92.91  82.85  10.06   2.20  R

R  Large residual
```

Response Surface Regression: Yield % versus Methanol to oil , Cat. loading (wt, Time (mins)

```
Analysis of Variance

Source                                          DF  Adj SS   Adj MS  F-Value  P-
Value
Model                                            5  3323.0   664.59    16.93
0.000
  Linear                                         3  2407.2   802.38    20.43
0.000
    Methanol to oil ratio                        1   409.6   409.56    10.43
0.004
    Cat. loading (wt.%)                          1  1880.7  1880.74    47.90
0.000
    Time (mins)                                  1   116.9   116.86     2.98
0.097
  Square                                         1   630.6   630.57    16.06
0.001
    Methanol to oil ratio*Methanol to oil ratio  1   630.6   630.57    16.06
0.001
  2-Way Interaction                              1   285.2   285.25     7.26
0.013
    Methanol to oil ratio*Time (mins)            1   285.2   285.25     7.26
0.013
Error                                           24   942.4    39.27
  Lack-of-Fit                                    7   476.7    68.11     2.49
0.059
  Pure Error                                    17   465.6    27.39
Total                                           29  4265.3


Model Summary

      S    R-sq  R-sq(adj)  R-sq(pred)
6.26623  77.91%     73.30%      65.69%


Coded Coefficients

Term                                          Effect    Coef  SE Coef  T-Value  P-
Value   VIF
Constant                                             69.97     1.67    41.78
0.000
Methanol to oil ratio                          10.12    5.06     1.57     3.23
0.004  1.00
Cat. loading (wt.%)                           -21.68  -10.84     1.57    -6.92
0.000  1.00
Time (mins)                                    -5.41   -2.70     1.57    -1.73
0.097  1.00
Methanol to oil ratio*Methanol to oil ratio    18.38    9.19     2.29     4.01
0.001  1.00
```

Methanol to oil ratio*Time (mins) -11.94 -5.97 2.22 -2.70 0.013 1.00

Regression Equation in Uncoded Units

Yield % = 90.6 - 4.60 Methanol to oil ratio - 10.84 Cat. loading (wt.%) + 0.616 Time (mins)
+ 1.021 Methanol to oil ratio*Methanol to oil ratio
- 0.1327 Methanol to oil ratio*Time (mins)

Fits and Diagnostics for Unusual Observations

Obs	Yield %	Fit	Resid	Std Resid	
29	82.14	69.97	12.17	2.02	R

R Large residual

RunOrder	PtType	Blocks	Methanol to oil ratio	Cat. loading (wt.%)	Time (mins)	Yield %
1	2	1	3	1	45	82.65
2	2	1	9	1	45	88.82
3	2	1	3	3	45	69.28
4	2	1	9	3	45	81.48
5	2	1	3	2	30	73.46
6	2	1	9	2	30	88.17
7	2	1	3	2	60	79.77
8	2	1	9	2	60	75.07
9	2	1	6	1	30	82.45
10	2	1	6	3	30	63.21
11	2	1	6	1	60	69.92
12	2	1	6	3	60	52.61
13	0	1	6	2	45	72.38
14	0	1	6	2	45	68.45
15	0	1	6	2	45	73.46
16	2	1	3	1	45	87.92
17	2	1	9	1	45	92.73
18	2	1	3	3	45	53.46
19	2	1	9	3	45	75.20
20	2	1	3	2	30	63.61
21	2	1	9	2	30	90.80
22	2	1	3	2	60	82.67
23	2	1	9	2	60	81.50
24	2	1	6	1	30	92.91
25	2	1	6	3	30	52.61
26	2	1	6	1	60	73.18
27	2	1	6	3	60	49.26
28	0	1	6	2	45	73.82
29	0	1	6	2	45	82.14
30	0	1	6	2	45	73.21

APPENDIX IV: Regression Equation (Uncoded Units):

$$\text{Yield \%} = 74.1 - 5.85X_1 - 8.8X_2 + 1.601X_3 + 0.966X_1^2 - 3.66X_2^2 - 0.01434X_3^2 + 0.957X_1X_2$$

$$-0.1327X_1X_3+0.153X_2X_3$$

Where: X_1 = Methanol to oil ratio, X_2 = Catalyst loading (wt.%), X_3 = Time (mins).

APPENDIX V: Regression Equation (Uncoded Units):

Regression Equation (Uncoded Units)

$$\text{Yield \%} = 90.6 - 4.60X_1 - 10.84X_2 + 0.616X_3 + 1.021X_1^2 - 0.1327X_1X_3$$

Where: X_1 = Methanol to oil ratio, X_2 = Catalyst loading (wt.%), X_3 = Time (mins).